Week 11 Manual Calculation Guide: Clustering

This guide is based on `week11.ipynb` and the class spreadsheet `K clustering calculation (230526).xlsx`.

Important exam habit: always use the values shown in the exam question. The notebook CSV and class spreadsheet have small differences in some example values, so do not memorize one table blindly.

1. What Week 11 Is Testing

Week 11 focuses on clustering, especially:

- K-means: numerical data, centroid is the mean.
- K-modes: categorical data, centroid is the mode.
- Euclidean distance: common distance for K-means.
- Manhattan distance: possible alternative distance.
- Reassignment step: assign each observation to the nearest cluster.
- Update step: recalculate the cluster center after assignment.

The manual calculation usually asks you to do one or more of these:

- calculate distance between two observations;
- assign observations to clusters;
- calculate new centroids for K-means;
- calculate dissimilarity scores for K-modes;
- calculate new modes for K-modes;
- repeat until cluster assignments no longer change.

2. Formula Sheet

Euclidean Distance

Use for numerical data.

For two observations:

Observation	x1	x2	x3
A	a1	a2	a3
B	b1	b2	b3

The Euclidean distance is:

```
d(A, B) = sqrt((a1 - b1)^2 + (a2 - b2)^2 + (a3 - b3)^2)
```

For 5 variables:

```
d(A, B) = sqrt((x1A - x1B)^2 + (x2A - x2B)^2 + (x3A - x3B)^2 + (x4A - x4B)^2 +
(x5A - x5B)^2)
```

Manhattan Distance

Use if the question specifically asks for Manhattan distance.

```
d(A, B) = |x1A - x1B| + |x2A - x2B| + |x3A - x3B| + ...
```

K-Means Centroid

For numerical variables, the centroid is the average of each variable inside the cluster.

Example for cluster A with 3 observations:

```
centroid x1 = (x1 of obs 1 + x1 of obs 2 + x1 of obs 3) / 3
centroid x2 = (x2 of obs 1 + x2 of obs 2 + x2 of obs 3) / 3
centroid x3 = (x3 of obs 1 + x3 of obs 2 + x3 of obs 3) / 3
```

Do this separately for every feature.

K-Modes Dissimilarity

Use for categorical data.

For each feature:

```
same category     = 0
different category = 1
```

Then add the scores.

Example:

Object	x1	x2	x3
O1	A	L	M
C1	A	L	C

```
x1: A vs A = 0
x2: L vs L = 0
x3: M vs C = 1
```

```
dissimilarity = 0 + 0 + 1 = 1
```

K-Modes Mode

For categorical variables, the cluster center is the most frequent category for each feature.

Example:

Object	x1	x2	x3
O1	A	L	M
O3	A	L	C
O5	A	P	M

Mode for each column:

```
x1: A, A, A -> A
x2: L, L, P -> L
x3: M, C, M -> M

new mode = (A, L, M)
```

3. Manual K-Means Procedure

Use this checklist in the exam.

1. Write down the initial centroids or initial cluster labels.
2. If initial labels are given, calculate the centroid of each cluster first.
3. For every observation, calculate distance to each centroid.
4. Assign the observation to the cluster with the smallest distance.
5. Recalculate the centroid of every cluster using the new members.
6. Repeat distance calculation and reassignment.
7. Stop when the cluster assignments do not change.

The key phrase is:

```
nearest centroid wins
```

4. Worked Example: Euclidean Distance

From the class spreadsheet, K-means Example 1 uses this data:

i	x1	x2	x3	x4	x5	Initial group
1	10	2	-1	4	0	A
2	12	4	-5	4	1	B
3	10	6	-6	4	0	C
4	9	2	-1	5	0	A
5	10	6	-3	4	0	B
6	9	4	-4	5	1	C
7	8	4	-5	5	1	A
8	10	6	-1	5	0	B

Distance Between Observation 1 and Observation 2

Observation 1:

```
(10, 2, -1, 4, 0)
```

Observation 2:

```
(12, 4, -5, 4, 1)
```

Step-by-step:

Feature	Calculation	Squared difference
x1	(10 - 12)^2	4
x2	(2 - 4)^2	4
x3	(-1 - -5)^2 = 4^2	16
x4	(4 - 4)^2	0
x5	(0 - 1)^2	1

```
sum = 4 + 4 + 16 + 0 + 1 = 25
distance = sqrt(25) = 5
```

So:

```
d(O1, O2) = 5
```

Distance Between Observation 1 and Observation 3

Observation 3:

```
(10, 6, -6, 4, 0)
```

```
d(O1, O3)
= sqrt((10 - 10)^2 + (2 - 6)^2 + (-1 - -6)^2 + (4 - 4)^2 + (0 - 0)^2)
= sqrt(0 + 16 + 25 + 0 + 0)
= sqrt(41)
= 6.403
```

Distance Between Observation 1 and Observation 4

Observation 4:

```
(9, 2, -1, 5, 0)
```

```
d(O1, O4)
= sqrt((10 - 9)^2 + (2 - 2)^2 + (-1 - -1)^2 + (4 - 5)^2 + (0 - 0)^2)
= sqrt(1 + 0 + 0 + 1 + 0)
= sqrt(2)
= 1.414
```

5. Worked Example: K-Means Centroid Calculation

Using the initial groups from Example 1:

```
A = O1, O4, O7
B = O2, O5, O8
C = O3, O6
```

Centroid A

Members:

i	x1	x2	x3	x4	x5
1	10	2	-1	4	0
4	9	2	-1	5	0
7	8	4	-5	5	1

Calculate average column by column:

```
x1 = (10 + 9 + 8) / 3 = 9
x2 = (2 + 2 + 4) / 3 = 2.667
x3 = (-1 + -1 + -5) / 3 = -2.333
x4 = (4 + 5 + 5) / 3 = 4.667
x5 = (0 + 0 + 1) / 3 = 0.333
```

So:

```
CA = (9, 2.667, -2.333, 4.667, 0.333)
```

Centroid B

Members:

```
B = O2, O5, O8
```

```
x1 = (12 + 10 + 10) / 3 = 10.667
x2 = (4 + 6 + 6) / 3 = 5.333
x3 = (-5 + -3 + -1) / 3 = -3
x4 = (4 + 4 + 5) / 3 = 4.333
x5 = (1 + 0 + 0) / 3 = 0.333
```

So:

```
CB = (10.667, 5.333, -3, 4.333, 0.333)
```

Centroid C

Members:

```
C = O3, O6
```

```
x1 = (10 + 9) / 2 = 9.5
x2 = (6 + 4) / 2 = 5
x3 = (-6 + -4) / 2 = -5
x4 = (4 + 5) / 2 = 4.5
x5 = (0 + 1) / 2 = 0.5
```

So:

```
CC = (9.5, 5, -5, 4.5, 0.5)
```

6. Worked Example: Full K-Means Reassignment

From the spreadsheet K-means Example 2:

i	x1	x2	x3	Initial group
1	10	2	-1	A
2	11	3	-1	B
3	18	5	-1	A
4	20	4	0	B
5	19	3	0	A
6	8	2	-1	B

Step 1: Calculate Initial Centroids

Initial groups:

```
A = O1, O3, O5
B = O2, O4, O6
```

Centroid A:

```
x1 = (10 + 18 + 19) / 3 = 15.667
x2 = (2 + 5 + 3) / 3 = 3.333
x3 = (-1 + -1 + 0) / 3 = -0.667

CA = (15.667, 3.333, -0.667)
```

Centroid B:

```
x1 = (11 + 20 + 8) / 3 = 13
x2 = (3 + 4 + 2) / 3 = 3
x3 = (-1 + 0 + -1) / 3 = -0.667

CB = (13, 3, -0.667)
```

Step 2: Calculate Distance to Both Centroids

Use:

```
d(Oi, CA) = sqrt((x1i - x1CA)^2 + (x2i - x2CA)^2 + (x3i - x3CA)^2)
```

i	d to CA	d to CB	New group
1	5.831	3.180	B
2	4.690	2.028	B
3	2.887	5.395	A
4	4.435	7.102	A
5	3.416	6.037	A
6	7.789	5.110	B

New groups:

```
A = O3, O4, O5
B = O1, O2, O6
```

Because the groups changed, recalculate the centroids.

Step 3: Recalculate Centroids

New centroid A:

```
A = O3, O4, O5

x1 = (18 + 20 + 19) / 3 = 19
x2 = (5 + 4 + 3) / 3 = 4
x3 = (-1 + 0 + 0) / 3 = -0.333

CA = (19, 4, -0.333)
```

New centroid B:

```
B = O1, O2, O6

x1 = (10 + 11 + 8) / 3 = 9.667
x2 = (2 + 3 + 2) / 3 = 2.333
x3 = (-1 + -1 + -1) / 3 = -1

CB = (9.667, 2.333, -1)
```

Step 4: Reassign Again

i	d to CA	d to CB	New group
1	9.244	0.471	B
2	8.090	1.491	B
3	1.563	8.750	A
4	1.054	10.515	A
5	1.054	9.410	A
6	11.200	1.700	B

The groups are still:

```
A = O3, O4, O5
B = O1, O2, O6
```

No change happened, so the K-means algorithm stops.

Final answer:

```
Cluster A: O3, O4, O5
Cluster B: O1, O2, O6
```

```
Final CA = (19, 4, -0.333)
Final CB = (9.667, 2.333, -1)
```

7. Manual K-Modes Procedure

Use K-modes when the variables are categorical, such as:

```
A, B, L, P, M, C
```

Procedure:

1. Write down the initial modes.
2. For each observation, compare it to every mode.
3. Put 0 if the category is the same.
4. Put 1 if the category is different.
5. Add the dissimilarity score.
6. Assign the observation to the cluster with the smallest score.
7. Recalculate the mode for each cluster.
8. Repeat until assignments do not change.

The key phrase is:

```
lowest dissimilarity wins
```

8. Worked Example: K-Modes Dissimilarity

From the spreadsheet:

i	x1	x2	x3
1	A	L	M
2	B	P	C
3	A	L	C
4	B	L	C
5	A	P	M

Initial modes:

```
C1 = (A, L, C)
C2 = (B, P, C)
```

Observation 1 Compared With C1

```
O1 = (A, L, M)
C1 = (A, L, C)
```

Feature	Compare	Score
x1	A vs A	0
x2	L vs L	0
x3	M vs C	1

```
d(O1, C1) = 0 + 0 + 1 = 1
```

Observation 1 Compared With C2

```
O1 = (A, L, M)
C2 = (B, P, C)
```

Feature	Compare	Score
x1	A vs B	1
x2	L vs P	1
x3	M vs C	1

```
d(O1, C2) = 1 + 1 + 1 = 3
```

Since 1 is smaller than 3:

```
O1 belongs to C1
```

Full First Assignment Table

i	Oi	d to C1	d to C2	Assigned cluster
1	(A, L, M)	1	3	C1
2	(B, P, C)	2	0	C2
3	(A, L, C)	0	2	C1
4	(B, L, C)	1	1	tie

i	Oi	d to C1	d to C2	Assigned cluster
5	(A, P, M)	2	2	tie

If there is a tie, follow the rule given by the lecturer or question. If no rule is given, state your tie decision clearly, for example:

```
Tie rule used: keep the existing cluster / choose the first cluster / use
lecturer's given assignment.
```

Do not silently hide a tie in your answer.

Recalculate K-Modes Centers

Suppose after assignment:

```
C1 = O1, O3, O4, O5
C2 = O2
```

For C1:

Object	x1	x2	x3
O1	A	L	M
O3	A	L	C
O4	B	L	C
O5	A	P	M

Mode column by column:

```
x1: A appears 3 times, B appears 1 time -> A
x2: L appears 3 times, P appears 1 time -> L
x3: M appears 2 times, C appears 2 times -> tie
```

If the question gives a tie rule, use it. If not, state the tie clearly.

Possible C1 mode:

```
C1 = (A, L, M) or (A, L, C), depending on tie rule for x3
```

For C2:

```
C2 = O2 = (B, P, C)
```

So:

```
C2 = (B, P, C)
```

9. K-Means vs K-Modes

Item	K-means	K-modes
Data type	Numerical	Categorical
Cluster center	Mean / centroid	Mode
Distance	Euclidean or Manhattan	Matching dissimilarity
Feature comparison	Squared difference or absolute difference	Same = 0, different = 1
Update step	Average each variable	Most frequent category

10. Exam Answer Templates

Template: Euclidean Distance

```
d(O__, O__) = sqrt((__ - __)^2 + (__ - __)^2 + (__ - __)^2)
            = sqrt(__ + __ + __)
            = sqrt(__)
            = __
```

For 5 variables:

```
d(O__, O__) = sqrt((x1 difference)^2 + (x2 difference)^2 + (x3 difference)^2 + (x4 difference)^2 + (x5 difference)^2)
```

Template: K-Means Centroid

```
Cluster A members = O__, O__, O__

CA_x1 = (__ + __ + __) / 3 = __
CA_x2 = (__ + __ + __) / 3 = __
CA_x3 = (__ + __ + __) / 3 = __

CA = (__, __, __)
```

Template: K-Means Assignment Table

Observation	d to CA	d to CB	Assigned cluster
O1			
O2			
O3			
O4			

Decision rule:

```
Choose the cluster with the smallest distance.
```

Template: K-Modes Dissimilarity

```
O__ = (__, __, __)
C__ = (__, __, __)

feature 1: same/different = __
feature 2: same/different = __
feature 3: same/different = __

dissimilarity = __ + __ + __ = __
```

Template: K-Modes Assignment Table

Observation	d to C1	d to C2	Assigned cluster
O1			
O2			
O3			
O4			

Decision rule:

```
Choose the cluster with the lowest dissimilarity score.
```

Template: K-Modes New Mode

```
Cluster C1 members = O__, O__, O__
```

```
x1 values = __, __, __ -> mode = __
x2 values = __, __, __ -> mode = __
x3 values = __, __, __ -> mode = __

New C1 = (__, __, __)
```

11. Common Mistakes To Avoid

- Do not average categorical values. Use mode for K-modes.
- Do not use mode for numerical K-means. Use mean.
- Do not forget the square root in Euclidean distance.
- Do not take the square root before adding all squared differences.
- Do not compare only one feature unless the question asks for it.
- Do not forget to recalculate centroids after reassignment.
- Do not stop K-means after one iteration unless the question says to do only one iteration.
- Do not ignore ties in K-modes. State the tie rule.
- Do not mix up "highest frequency" and "lowest dissimilarity": mode uses highest frequency, assignment uses lowest dissimilarity.

12. Quick Final-Exam Flow

When you see numerical clustering:

```
K-means -> calculate mean centroids -> calculate distance -> assign nearest -> update mean -> repeat
```

When you see categorical clustering:

```
K-modes -> compare same/different -> assign lowest dissimilarity -> update mode -> repeat
```

When you see mixed numerical and categorical data:

```
K-prototypes -> numerical part behaves like K-means, categorical part behaves like K-modes
```

For Week 11 manual calculation, the most important skills are:

```
Euclidean distance
K-means centroid update
K-modes dissimilarity score
K-modes mode update
```